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Math 3108 – Linear Algebra
Exam 1 – Part A
Prof. Runnion
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Directions:

- 1) All cell phones and other electronic noisemaking devices must be turned off or completely silenced (i.e. not on vibrate) for the duration of the exam.
- 2) No calculators, books, or other materials are permitted.
- 3) Show **ALL** your work! Correct answers which are not properly justified will not receive full credit.
- 4) Failure to follow directions specific to a problem will result in the loss of points.
- 5) Circle, box, or underline each final answer if necessary for clarity. Label all graphs.
- 6) If you work a problem two different ways, clearly indicate which one you want us to grade, preferably by crossing out the one you do not want us to look at. If you use two methods, one of which is wrong, and neither method is crossed out, you will not receive full credit.
- 7) Answers must be exact (like $\sqrt{2}$), not approximate (like 1.414), unless a problem specifically indicates otherwise.
- 8) Simplify where appropriate. Quantities such as $\sqrt{9}$ and $\cos \pi$ should be calculated.
- 9) If you need extra room, you may use the back of the previous page. However, you must indicate you are doing so by clearly writing "BPP" on the relevant problem.
- 10) This packet has four sheets of paper, including this cover page. Part B has three sheets of paper, including a cover page. Do NOT remove the staple or remove any sheet from this packet.
- 11) Once this exam begins, you will have 50 minutes to complete your solutions.

Page	1	2	3	Part A	Part B	Sum
Points Earned	14	20	22	56	42	98
Maximum Points	14	20	22	56	44	100

**DO NOT OPEN THIS EXAM
UNTIL TOLD TO DO SO**

Name: _____

Multiple Choice and True/False. Circle the correct answer.

1. (8 points) Consider the matrix $A = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

i) A is in _____.

(circle one)

Echelon Form

Reduced Echelon Form

Neither

ii) The equation $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution.

(circle one)

True

False

x_3 is free

iii) The transformation $T(\mathbf{x}) = A\mathbf{x}$ is _____ and _____.

(circle one)

One-to-One

Not One-to-One

(circle one)

Onto

Not Onto

2. (2 points) If $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^5$, then the vector $-\mathbf{u}$ is in _____.

a) $\{\mathbf{u}, \mathbf{v}\}$

b)

Span $\{\mathbf{u}, \mathbf{v}\}$

c) Both (a) and (b)

d)

Neither (a) nor (b)

3. (2 points) Let $A = \begin{bmatrix} -2 & 1 \\ 3 & 5 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$. Which of the following quantities produces a 2×2 matrix? (Circle all that apply)

a) $(A\mathbf{x})^T$

b)

$\mathbf{x}^T A^T$

c)

$A^T \mathbf{x}^T$

d)

$\mathbf{x}\mathbf{x}^T$

e)

$\mathbf{x}^T \mathbf{x}$

f)

None of these

4. (2 points) Let $A = \begin{bmatrix} -2 & 1 \\ 3 & 5 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$. Which of the following quantities is undefined? (Circle all that apply)

a) $(A\mathbf{x})^T$

b)

$\mathbf{x}^T A^T$

c)

$A^T \mathbf{x}^T$

d)

$\mathbf{x}\mathbf{x}^T$

e)

$\mathbf{x}^T \mathbf{x}$

f)

None of these

Name: _____

Open Answer. Show ALL your work!

5. For this problem, consider the matrix $A = [a_1 \ a_2 \ a_3] = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 2 & 1 \\ 3 & 1 & 1 \end{bmatrix}$, which can be row

reduced into the echelon form $\begin{bmatrix} 1 & 3 & 1 \\ 0 & -4 & -1 \\ 0 & 0 & 0 \end{bmatrix}$. (You do not need to verify this.)

- a) (5 points) Do the columns of A span $\mathbb{R}^3$? Justify your answer using words.

5 No, Because The Coefficient Matrix A
Does not Have a pivot in Every Row

- b) (5 points) Is $v = \begin{bmatrix} 9 \\ 6 \\ 3 \end{bmatrix}$ in $\text{Span}\{a_1, a_2, a_3\}$? If so, show how v can be written as a linear combination of a_1 , a_2 , and a_3 .

5 Yes ; $v = 3(\vec{a}_2) = 3 \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 9 \\ 6 \\ 3 \end{bmatrix}$

6. (10 points) When propane gas burns, the propane (C_3H_8) combines with oxygen (O_2) to form carbon dioxide (CO_2) and water (H_2O) according to an equation of the form



10 Set up an appropriate vector equation which could be used to balance this chemical equation. There is no need to actually solve the vector equation. Be sure that you clearly indicate what the entries in the vectors represent.

$$\begin{bmatrix} C \\ H \\ O \end{bmatrix} \rightarrow \left/ x_1 \begin{bmatrix} 3 \\ 8 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + x_4 \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} \right/$$

$$\rightarrow \left[\begin{array}{cccc|c} 3 & 0 & -1 & 0 & 0 \\ 8 & 0 & 0 & -2 & 0 \\ 0 & 2 & -2 & -1 & 0 \end{array} \right]$$

Name: _____

7. a) (14 points) Solve the matrix equation $A\mathbf{x} = \mathbf{b}$ using augmented matrices, where

$$A = \begin{bmatrix} 1 & 2 & -4 \\ 0 & 1 & -3 \\ 2 & 1 & 1 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 3 \\ -1 \\ 9 \end{bmatrix}.$$

Show all your work! State your solution in parametric vector form.

$$\begin{array}{l} \rightarrow -2R_1 + R_3 \rightarrow R_3 \\ \rightarrow 3R_2 + R_3 \rightarrow R_3 \end{array} \left[\begin{array}{ccc|c} 1 & 2 & -4 & 3 \\ 0 & 1 & -3 & -1 \\ 0 & -3 & 9 & 3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & -4 & 3 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\rightarrow -2R_2 + R_1 \rightarrow R_1 \left[\begin{array}{ccc|c} \textcircled{1} & 0 & 2 & 5 \\ 0 & \textcircled{1} & -3 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right] \left. \begin{array}{l} x_1 + 2x_3 = 5 \\ x_2 - 3x_3 = -1 \\ x_3 \text{ is Free} \end{array} \right\}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2x_3 + 5 \\ 3x_3 - 1 \\ x_3 \end{bmatrix}$$

14

$$\boxed{\vec{x} = x_3 \begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix} + \begin{bmatrix} 5 \\ -1 \\ 0 \end{bmatrix}} \quad ; \quad \vec{x} = \vec{v}_h + \vec{p}$$

- b) (4 points) Without performing any additional row operations, state the solution set for the homogeneous equation $A\mathbf{x} = \mathbf{0}$ where A is the matrix given in part (a) above.

4

$$\vec{x} = \vec{v}_h \quad ; \quad \boxed{\vec{x} = x_3 \begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix}}$$

8. (4 points) Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation where $T(\mathbf{u}) = \begin{bmatrix} 4 \\ -2 \end{bmatrix}$ and

$$T(\mathbf{v}) = \begin{bmatrix} -1 \\ 3 \end{bmatrix}. \text{ Use the fact that } T \text{ is linear to find } T(\mathbf{u} - 2\mathbf{v}).$$

4

$$\begin{aligned} T(\mathbf{u} - 2\mathbf{v}) &= T(\mathbf{u}) - 2T(\mathbf{v}) = \begin{bmatrix} 4 \\ -2 \end{bmatrix} - 2 \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \end{bmatrix} - \begin{bmatrix} -2 \\ 6 \end{bmatrix} \\ &= \begin{bmatrix} 6 \\ -8 \end{bmatrix} \end{aligned}$$